DO NOW — RECALL

# Boolean algebra rules

Name: _____ Class: _____ Date: _____

**1. Normalise $0.0110 \times 2^4$ using a two’s-complement mantissa. Give the new mantissa and exponent.**

**2. A sorted array is [3, 8, 12, 19, 27, 35, 42] . When searching for 27 , which indexes are checked, in order?**

**3. Show the array after one full left-to-right pass of ascending bubble sort on [7, 3, 5, 2] .**

**4. An alarm sounds if the door is open AND the system is armed, OR if panic is pressed. Write Q using A, B and C.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

0.1100 × $2^3$ . Shift the mantissa one place left and decrease the exponent by one to preserve the value. A normalised positive two’s-complement mantissa begins 01 ; a negative one begins 10 .

### 2. Model answer

Indexes 3 → 5 → 4 (values 19 → 35 → 27). Binary search tests the middle index of the remaining sorted range, using zero-based indexing.

### 3. Model answer

[3, 5, 2, 7] . Compare adjacent values and swap when the left value is larger. The largest value, 7, moves to the end.

### 4. Model answer

Q = (A ∧ B) ∨ C . A = door open, B = armed, C = panic pressed. Group door AND armed first; panic is an alternative trigger.

### By the end, you can…

• apply De Morgan's Laws and double negation

• use distribution, association and commutation to rearrange expressions

• show each algebra step clearly in OCR-style simplification

### Vocabulary — know these terms

Recall the term — De Morgan

Recall the term — distribution

Recall the term — association

Recall the term — commutation

Recall the term — double negation

Recall the term — simplify

Recall the term — equivalent

Recall the term — brackets

NEW LEARNING

# Read, annotate and explain

## Change form without changing truth

De Morgan:
¬(A ∧ B) = ¬A ∨ ¬B
¬(A ∨ B) = ¬A ∧ ¬B

Double negation:
¬¬A = A

Name the rule, apply one change at a time, and keep the expression logically equivalent.

## De Morgan

The operator inside the brackets swaps as each term is negated.

## Double negation

Two consecutive NOT operations cancel.

## Distributive law

A appears in both groups. Write it once outside the brackets, then join B and C with OR.

The matching OR form is also useful: (A ∨ B) ∧ (A ∨ C) = A ∨ (B ∧ C).

## Association + commutation

These allow regrouping and reordering before other simplification.

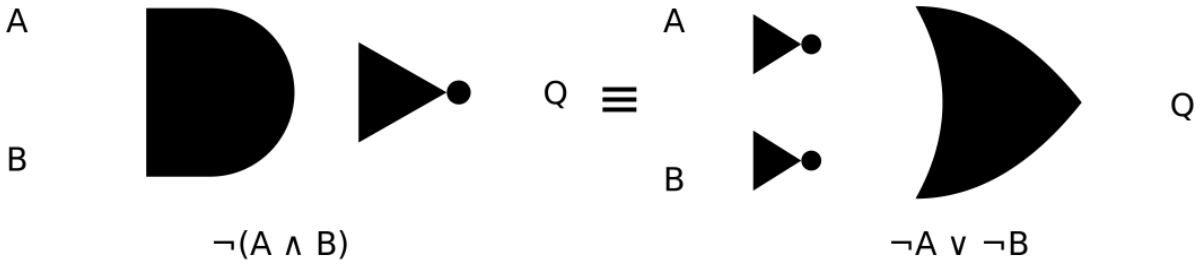
**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### De Morgan

The operator inside the brackets swaps as each term is negated.




Two equivalent circuits: AND then NOT; and NOT A, NOT B, then OR

### Double negation

Two consecutive NOT operations cancel.



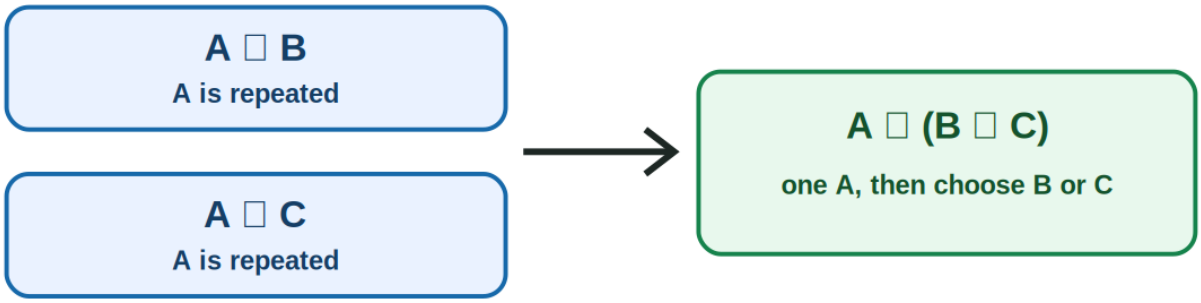

Input A passes through two NOT gates and returns to A

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### Distributive law

A appears in both groups. Write it once outside the brackets, then join B and C with OR.




Common factor diagram: A AND B, or A AND C, becomes A AND either B or C

I DO — WORKED EXAMPLE

# Follow the worked method

## Simplify ¬(A ∧ B)

## Outer NOT over AND

Use De Morgan's Law.

## Negate each term

A → ¬A
B → ¬B

## AND becomes OR

¬A ∨ ¬B

## Equivalent behaviour

Both are false only when A=1 and B=1.

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Simplify ¬(A ∧ B)**

Apply one named rule.




A and B enter an AND gate, which then connects to a NOT gate and output Q

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

### 1. De Morgan

Simplify ¬A ∨ ¬B using the relevant De Morgan form.

```
¬A ∨ ¬B
= ¬(? ? ?)
```

Method / working:

### 2. Double negation

Simplify ¬(¬B).

```
¬(¬B)
= ?
```

Method / working:

### 3. Distribution

Simplify (A ∨ B) ∧ (A ∨ C).

```
(A ∨ B) ∧ (A ∨ C)
= A ∨ (? ? ?)
```

Method / working:

### 4. Commutation + association

Rewrite C ∨ (A ∨ B) into (A ∨ B) ∨ C.

```
C ∨ (A ∨ B)
= ?
```

Method / working:

# WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

### Guided 1: De Morgan Simplify ¬A ∨ ¬B using the relevant De Morgan form. Complete the same expression CODE: ¬A ∨ ¬B = ¬(? ? ?)

DISCUSS — Compare approaches, walk through one test input, and explain how you would detect or correct a mistake.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

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### Guided 2: Double negation Simplify ¬(¬B).

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

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### Guided 3: Distribution Simplify (A ∨ B) ∧ (A ∨ C).

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

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### Guided 4: Commutation + association Rewrite C ∨ (A ∨ B) into (A ∨ B) ∨ C.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

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YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

## Question 1

Simplify Boolean expressions
State the simplified form of each expression:
(i) ¬¬A (ii) ¬A ∧ ¬B (iii) ¬(¬A ∧ ¬B)

## Question 2

Simplify the popcorn expression
A cinema gives free popcorn when a voucher is shown and either a soft drink or a bottle of water is purchased.
E = voucher shown; F = soft drink bought; G = water bought.
Simplify (E ∧ F) ∨ (E ∧ G) .

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Cinema discount: Boolean expression
A discounted cinema ticket is issued if the customer is under 18 AND has a student card , OR is over 60 AND has proof of age .
A = under 18; B = student card; C = over 60; D = ID. Write the Boolean expression for the output Q (discount ticket).

### Question 4

From a gate circuit to an expression
Two OR gates take inputs A, B and C, D respectively. Their outputs feed a third OR gate. Write the complete Boolean expression for the circuit output.

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**De Morgan's Laws swap AND and OR while negating each term.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

## GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

# WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

### Guided 1 — original CHECK

Complete the same expression

CODE:

¬A ∨ ¬B

= ¬(A ∧ B)

IMPROVE: Explain one correction to your own reasoning.

### Guided 2 — original CHECK

Complete the same expression

CODE:

¬(¬B)

= ?

CODE:

¬(¬B)

= B

IMPROVE: Explain one correction to your own reasoning.

### Guided 3 — original CHECK

Complete the same expression

CODE:

(A ∨ B) ∧ (A ∨ C)

= A ∨ (? ? ?)

CODE:

(A ∨ B) ∧ (A ∨ C)

= A ∨ (B ∧ C)

IMPROVE: Explain one correction to your own reasoning.

### Guided 4 — original CHECK

Complete the same expression

CODE:

C ∨ (A ∨ B)

= ?

CODE:

C ∨ (A ∨ B)

= (A ∨ B) ∨ C

IMPROVE: Explain one correction to your own reasoning.